CS 305: Computer Networks

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Lecture 5： Application Layer

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Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

2.6 video streaming and content distribution networks

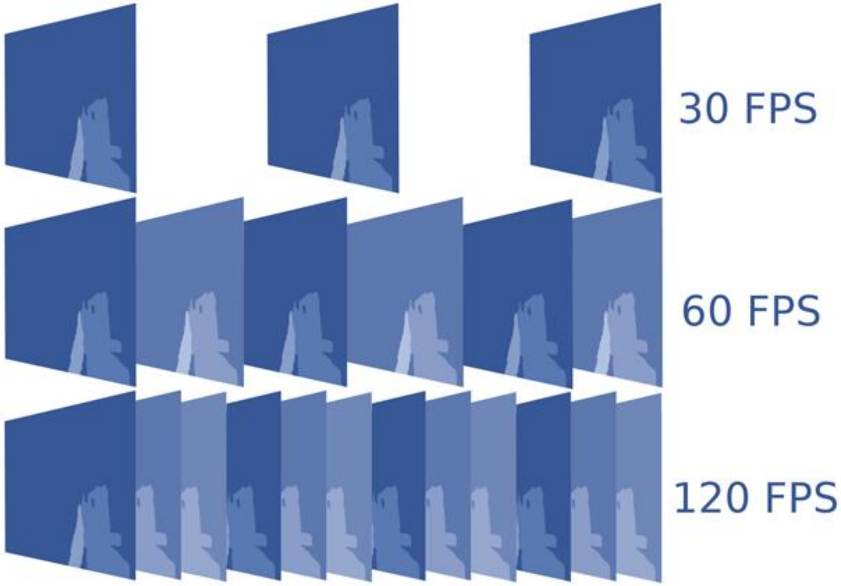
2.7 socket programming with UDP and TCP

Video Streaming and CDNs: ~~context~~

- Video traffic: major consumer of Internet bandwidth
  - Netflix, YouTube: 37%, 16% of downstream residential ISP traffic
  - ~1B YouTube users, ~75M Netflix users
- Challenge: scale - how to reach ~1B users?
  - single mega-video server won't work (why?)
- Challenge: heterogeneity
  - different users have different capabilities (e.g., wired versus mobile; bandwidth rich versus bandwidth poor)
- Solution: distributed, application-level infrastructure

Multimedia: video

- Video: sequence of images displayed at constant rate
  - e.g., 24 images/sec
- Digital image: array of pixels
  - each pixel represented by bits



Multimedia: video

Coding (Compression): use redundancy *within* and *between* images to decrease # bits used to encode image

- spatial (within image)
- temporal (from one image to next)

Spatial coding example: instead of sending N values of same color (all purple), send only two values: color value (purple) and number of repeated values (N)

frame *i*

frame *i+1*

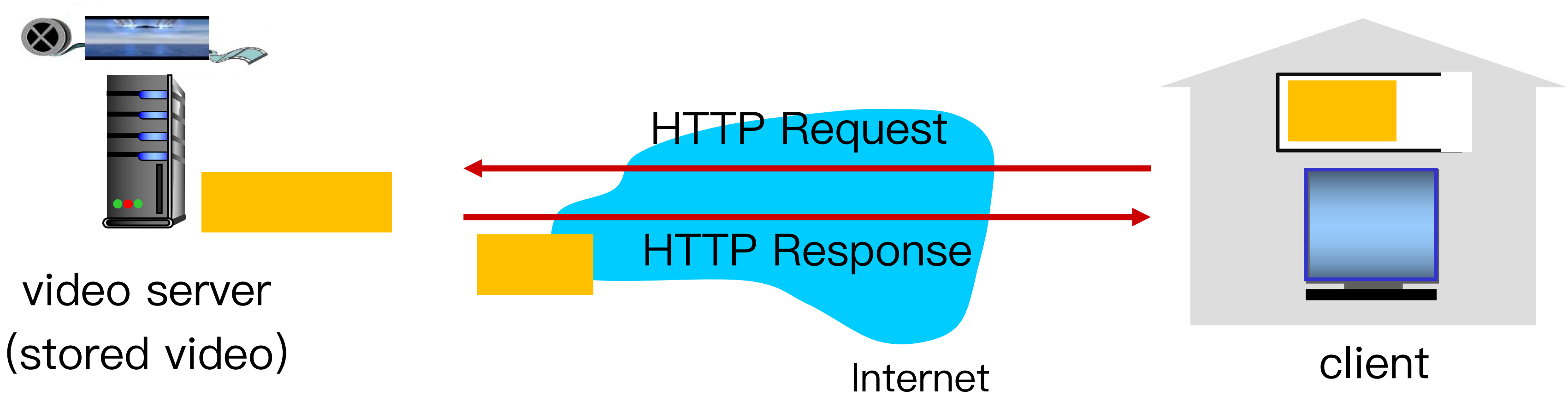
temporal coding example: instead of sending complete frame at i+1, send only differences from frame i

Multimedia: video

Type	Video Bitrate, Standard Frame Rate (24, 25, 30)	Video Bitrate, High Frame Rate (48, 50, 60)
2160p (4k)	35-45 Mbps	53-68 Mbps
1440p (2k)	16 Mbps	24 Mbps
1080p	8 Mbps	12 Mbps
720p	5 Mbps	7.5 Mbps
480p	2.5 Mbps	4 Mbps
360p	1 Mbps	1.5 Mbps

- CBR: (constant bit rate): video encoding rate fixed
- VBR: (variable bit rate): video encoding rate changes as amount of spatial, temporal coding changes

HTTP Streaming



All clients receive the same encoding of the video:

- Human users may have different requirements
- Clients may have different available bandwidth, which may be time-varying

How to deal with this?

Streaming multimedia: DASH

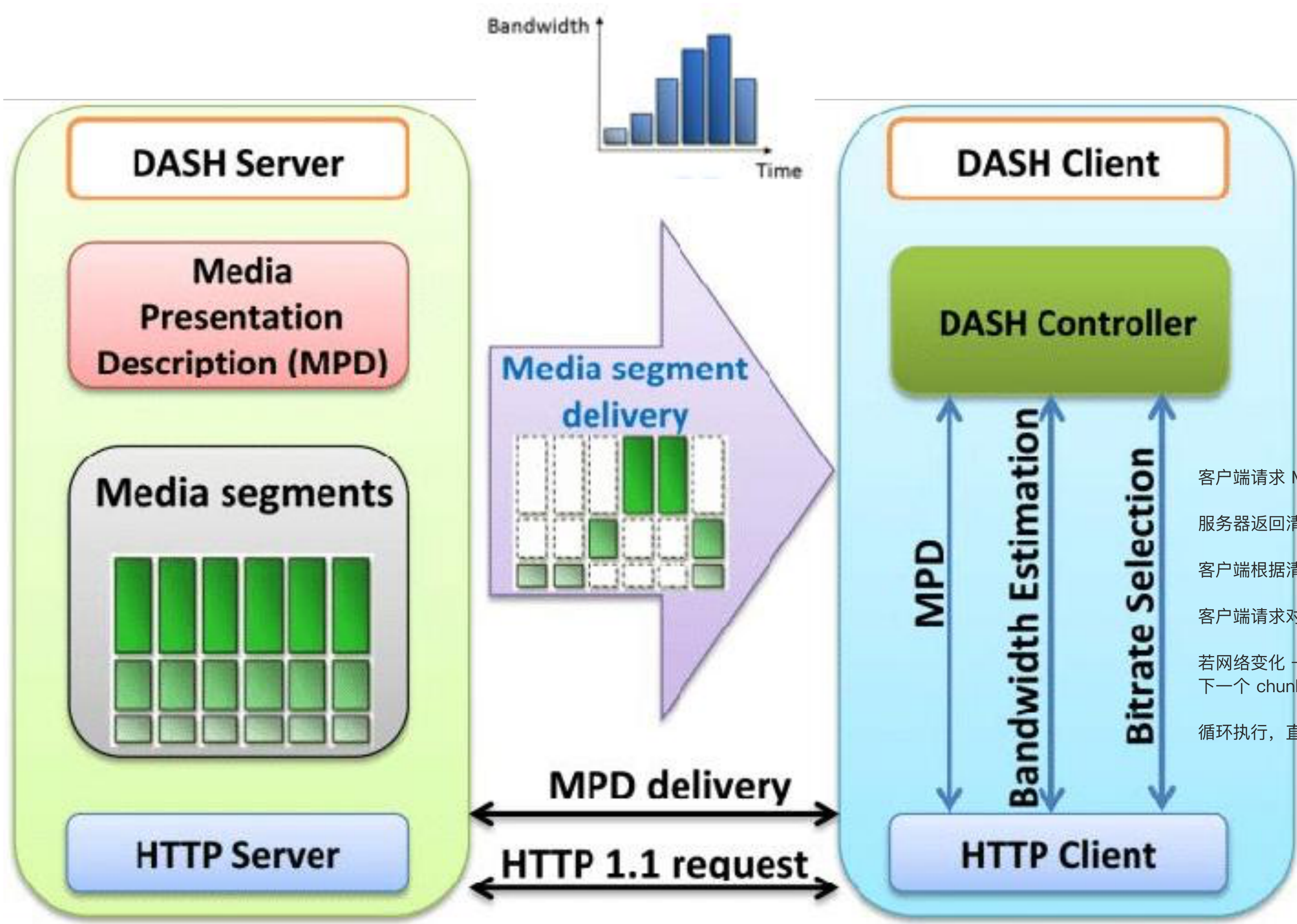
- DASH: Dynamic, Adaptive Streaming over HTTP
- Server:
  - divides video file into multiple chunks
  - each chunk stored, encoded at different rates
    每个 chunk 被编码成多个版本（不同清晰度和码率）：
  - manifest file: provides URLs for different chunks encoded at different rates
- Client:
  - periodically measures server-to-client bandwidth
  - consulting manifest, requests one chunk at a time
    - chooses maximum coding rate sustainable given current bandwidth 从 Manifest 文件中选择当前带宽能支持的最高码率版本；
    - can choose different coding rates at different points in time (depending on available bandwidth at time)

每个 chunk 请求都是独立的；

所以客户端可以在每个 chunk 之间切换清晰度；

实现“自适应流”（Adaptive Streaming）。

Streaming multimedia: DASH



客户端请求 MPD 文件；

服务器返回清单；

客户端根据清单选择适当码率；

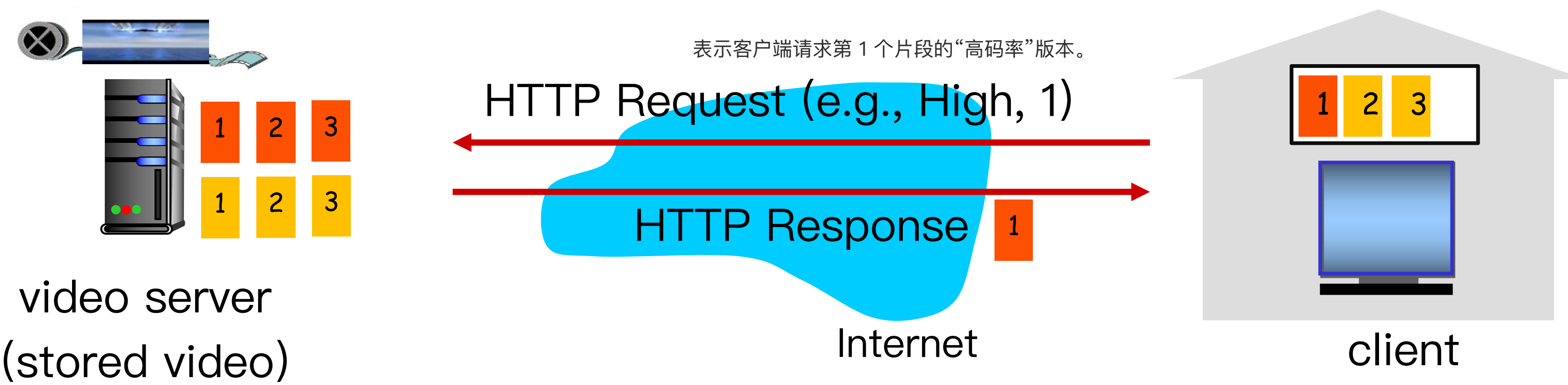
客户端请求对应 chunk；

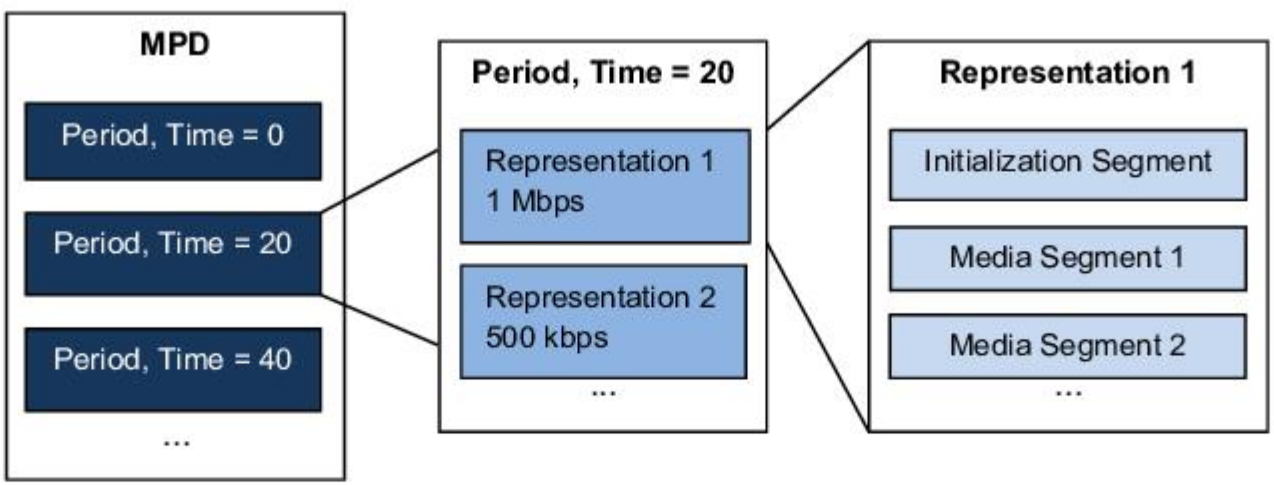
若网络变化 → 重新评估带宽 → 下一个 chunk 换更合适的码率；

循环执行，直到播放结束。

Streaming multimedia: DASH

表示客户端请求第 1 个片段的“高码率”版本。





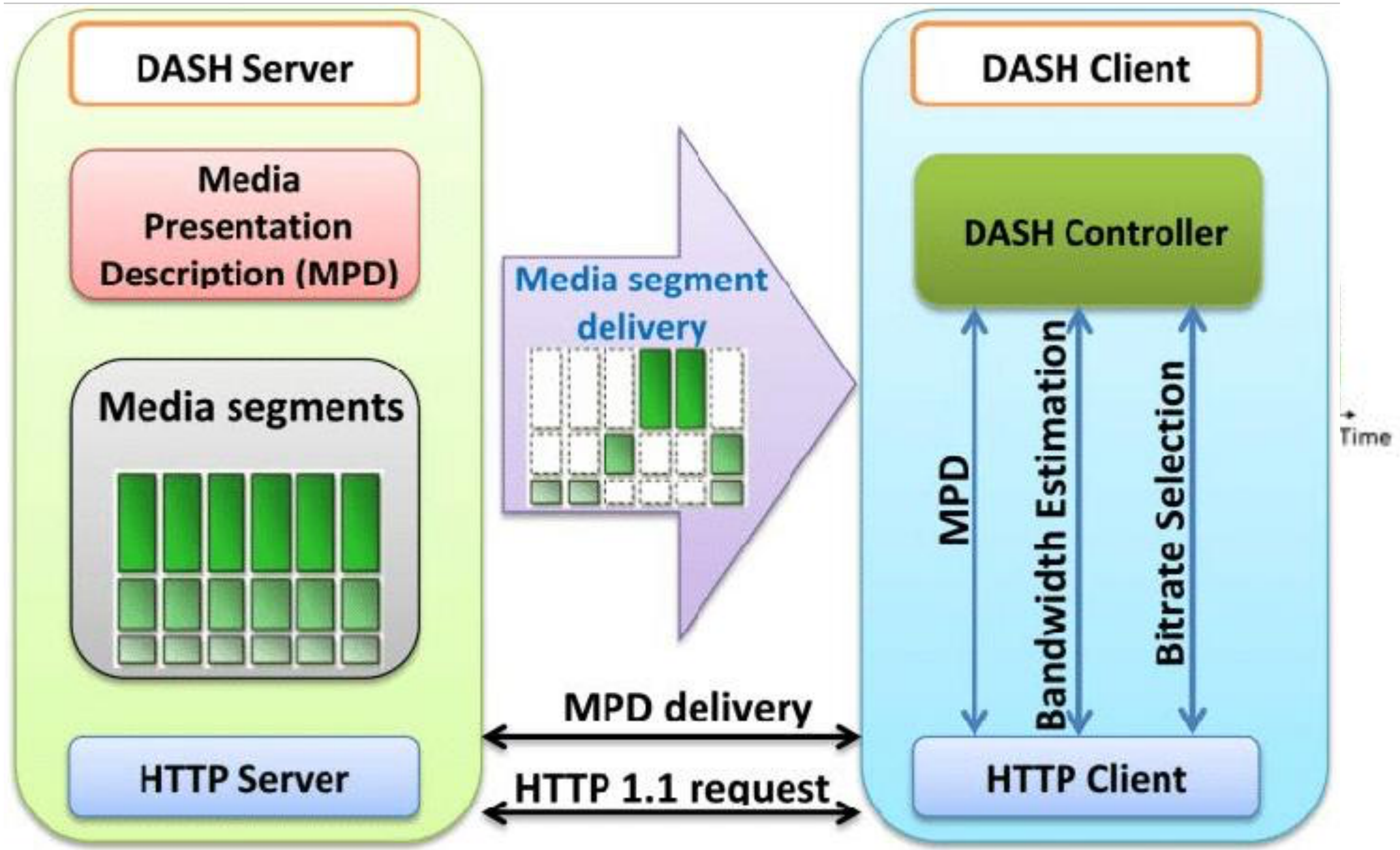
每个 Representation 里又包括：

Initialization Segment（初始化信息）

Media Segment 1, 2, …（真正的视频片段）

Media presentation description (MPD), also known as manifest

Streaming multimedia: DASH



Streaming multimedia: DASH

"intelligence" at client: client determines

在合适的时间点请求下一个 chunk；

- when to request chunk (so that buffer starvation, or overflow does not occur)

  客户端需要控制**缓冲区（buffer）**的大小：

  不能让 buffer 空了（starvation） → 会导致播放中断；
  也不能让 buffer 满了（overflow） → 浪费带宽和内存

- what encoding rate to request (higher quality when more bandwidth available)
- where to request chunk (can request from URL server that is "close" to client or has high available bandwidth)

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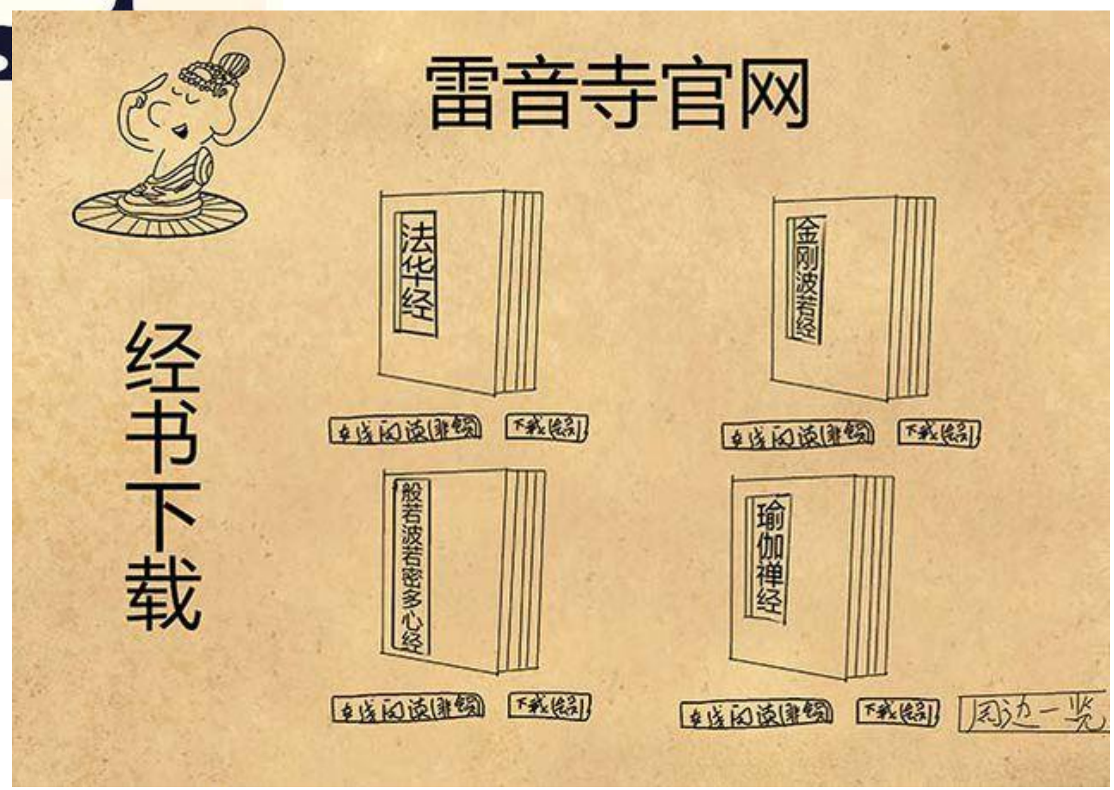
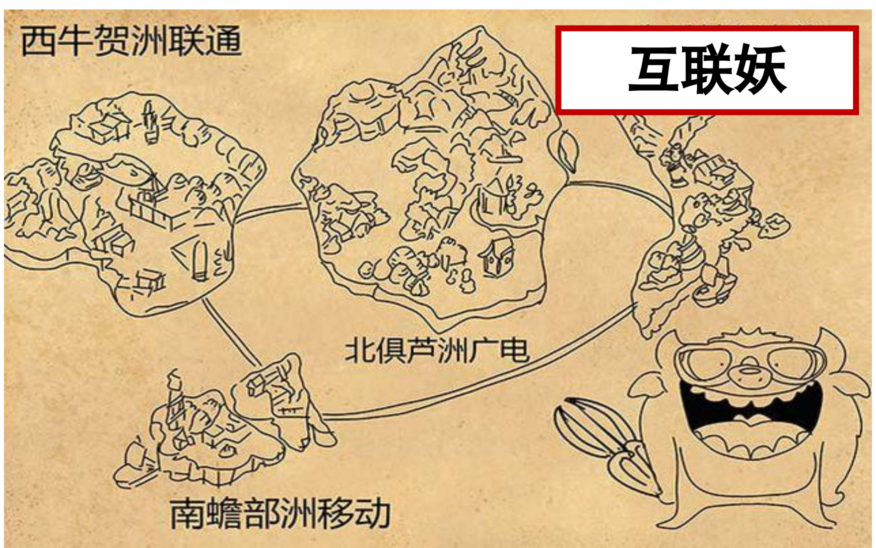
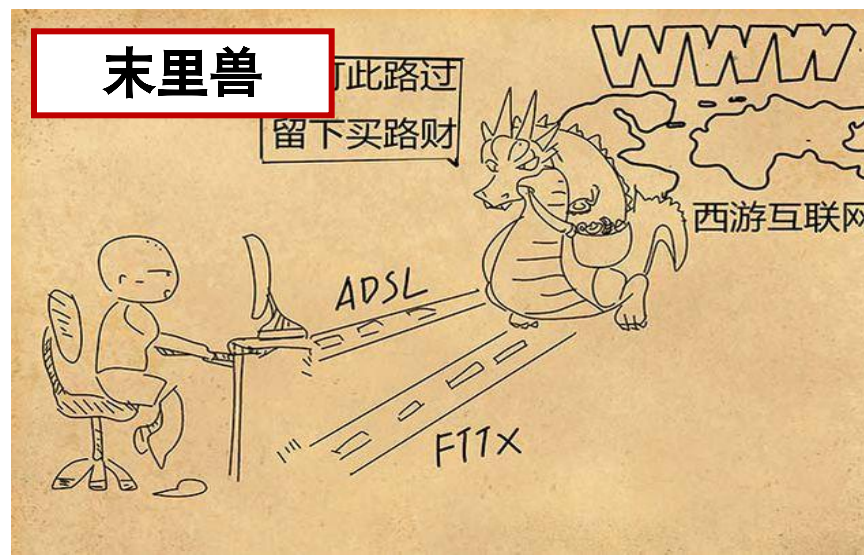
Content distribution networks



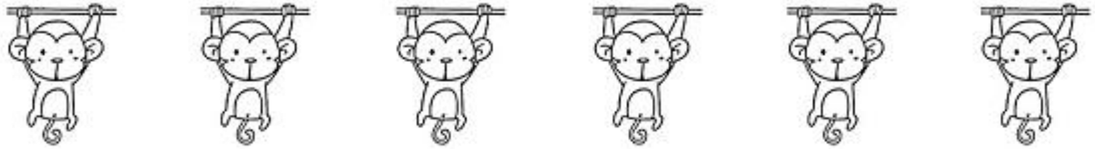
来源：小黑羊JoinWings
https://baijia.baidu.com/s?old_id=126615&wfr=pc&fr=app_list

Content distribution networks





Content distribution networks



Content distribution networks

- Challenge: how to stream content (selected from millions of videos) to hundreds of thousands of simultaneous users?

- Option 1: single, large "mega-server"
  - single point of failure
  - huge traffic
  - long path to distant clients
  - multiple copies of video sent over outgoing link

同一个视频可能被传输上百万次；

….quite simply: this solution doesn't scale

Content distribution networks

- Challenge: how to stream content (selected from millions of videos) to hundreds of thousands of simultaneous users?

- Option 2: Content distribution networks (CDN) store/serve multiple copies of videos at multiple geographically distributed sites
  - Enter deep: push CDN servers deep into many access networks; inside ISPs
    - close to users
    - used by Akamai, 1700 locations
  - Bring home: smaller number (10’s) of larger clusters in Internet Exchange Point (IXP); outside ISPs
    - used by Limelight

把 CDN 服务器“深入”部署到各地网络内部，比如放进 ISP（互联网服务提供商）网络。

离用户非常近，延迟最低。

在全球选取少量的大型集群节点；

这些节点部署在 Internet Exchange Points (IXP)，即互联网骨干网的交汇处；

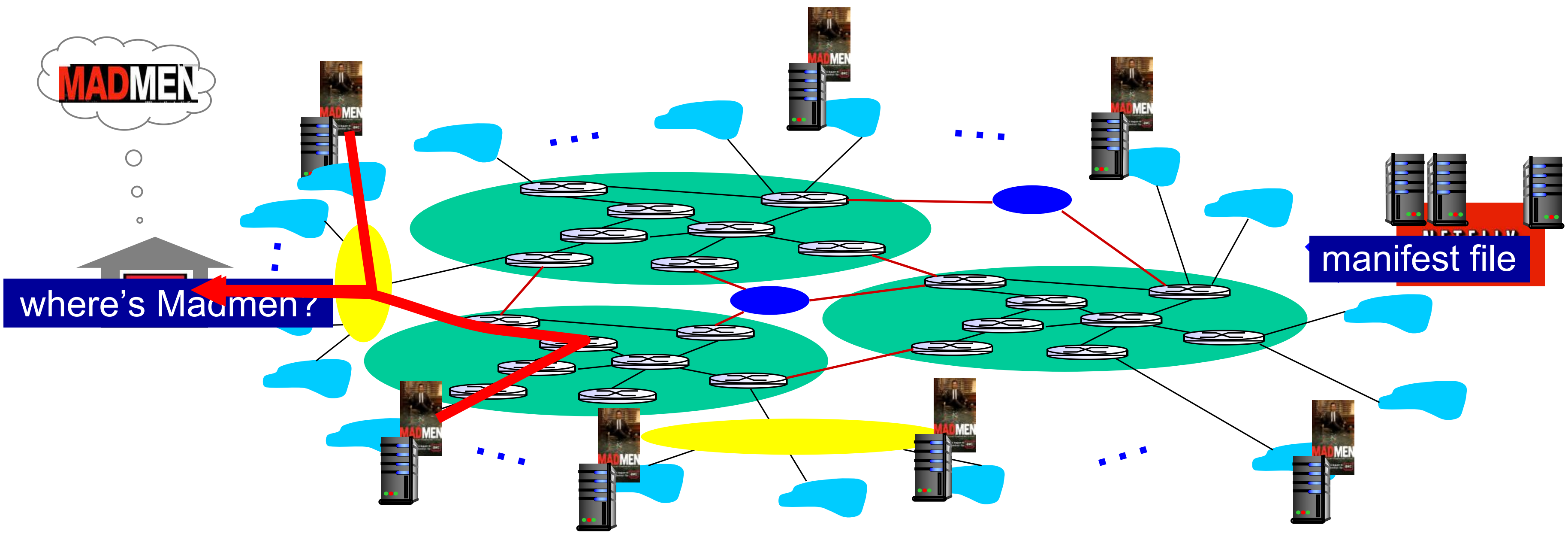
不直接进到 ISP 内部，而是在外部高速连接点上。

Content Distribution Networks (CDNs)

- CDN: stores copies of content at CDN nodes
  - e.g. Netflix stores copies of MadMen
- subscriber requests content from CDN
  - directed to nearby copy, retrieves content
  - may choose different copy if network path congested

每个 CDN 节点会保存某些视频或网页的副本；

不同用户实际上访问的是不同服务器上的副本内容，而不是同一个中央服务器。



Content Distribution Networks (CDNs)

Challenges: Coping with a congested Internet

- what content to place in CDN node?
  - Simple pull strategy: request, then store
- from which CDN node to retrieve content?
  - Cluster selection strategy
- the operation for retrieving content?

CDN 节点的存储空间有限，不可能保存所有视频。
所以需要决定哪些内容该放进去。

Simple Pull Strategy（简单拉取策略）

当有用户请求某个内容时，节点从主服务器下载该内容；

并在本地缓存，以便下次请求时直接提供；

类似浏览器的缓存机制：“request, then store”。

优点：节省空间，只缓存热门内容；
缺点：首次访问时延迟高（因为要去主服务器拉取）。

Cluster Selection Strategy（集群选择策略）。
当同一个内容存在多个节点时，CDN 要决定用户应连接哪个节点。

选择依据包括：

地理位置（哪个节点离用户近）；

网络延迟/带宽（哪个节点更快）；

负载情况（哪个节点更空闲）；

实时网络状态（避免拥堵路径）。

内容检索与缓存更新策略：

如果请求的内容在本地缓存中：

直接提供给用户；

如果本地没有：

从上级 CDN 节点或主服务器（origin server）拉取；

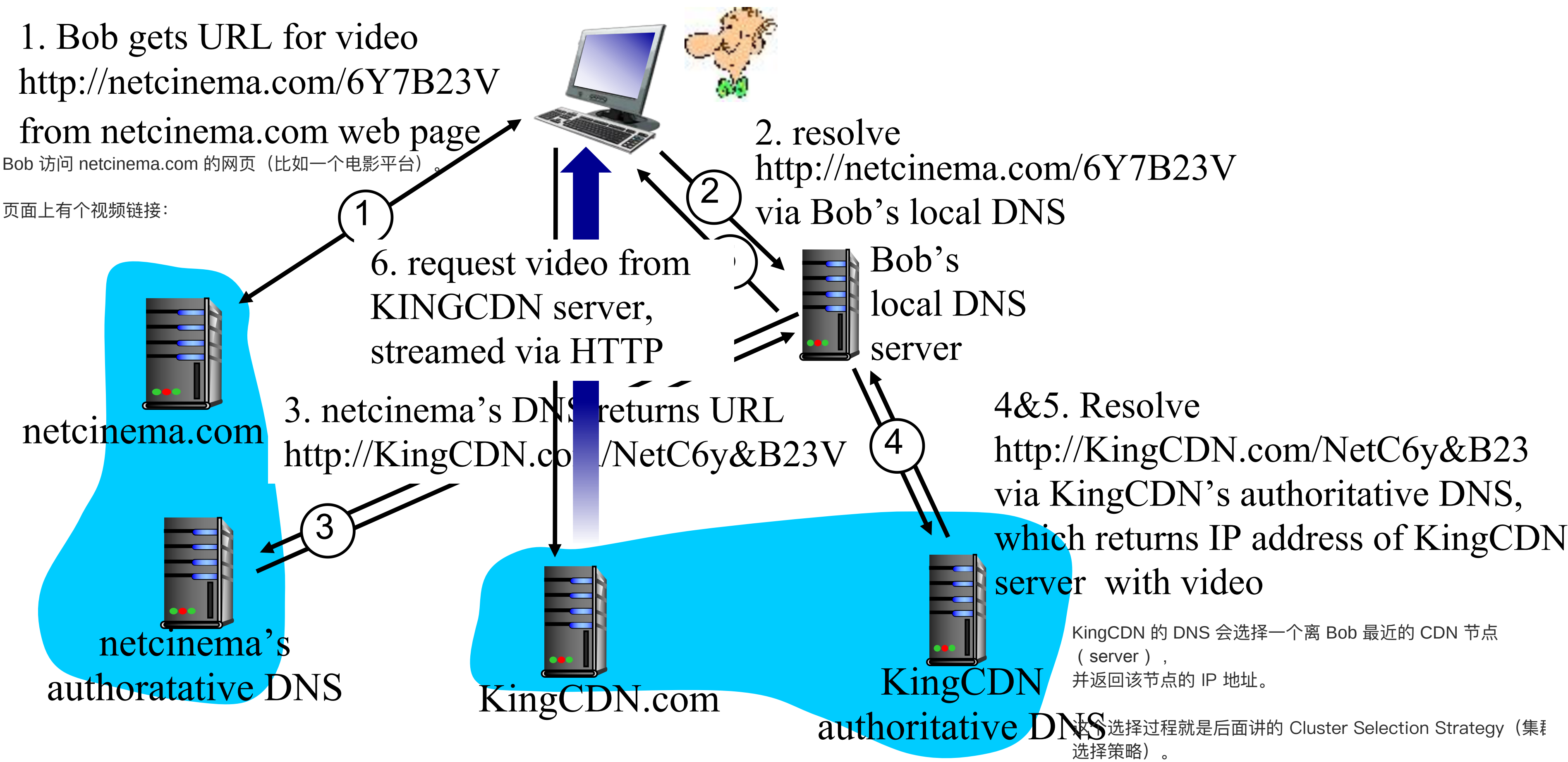
然后缓存一份副本。

这样 CDN 节点之间形成一个分层缓存体系（hierarchical cache structure）。

CDN Operation

Bob (client) requests video http://netcinema.com/6Y7B23V

- video stored in CDN at http://KingCDN.com/NetC6y&B23V



CDN: Cluster Selection Strategy

One simple strategy is to assign the client to the cluster that is **geographically closest**:

- When a DNS request is received from a particular LDNS, the CDN chooses the geographically closest cluster
- may not be the closest cluster in terms of the length or number of hops
- ignore the variation in delay and available bandwidth over time

Periodic **real-time measurements** of delay and loss performance between their clusters and clients:

- a CDN can have each of its clusters periodically send probes to all of the LDNSs around the world.
- many LDNSs are configured to not respond to such probes.

各 CDN 节点会周期性地向不同地区的本地 DNS 发出探测包（probe）。

记录响应时间，用以动态优化路由。

并不是所有的本地 DNS 都会响应这种探测；

所以有时候测量结果不完整。

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